

Z is for Zoning: Market Tightness and the Price Effects of Land Use Regulation

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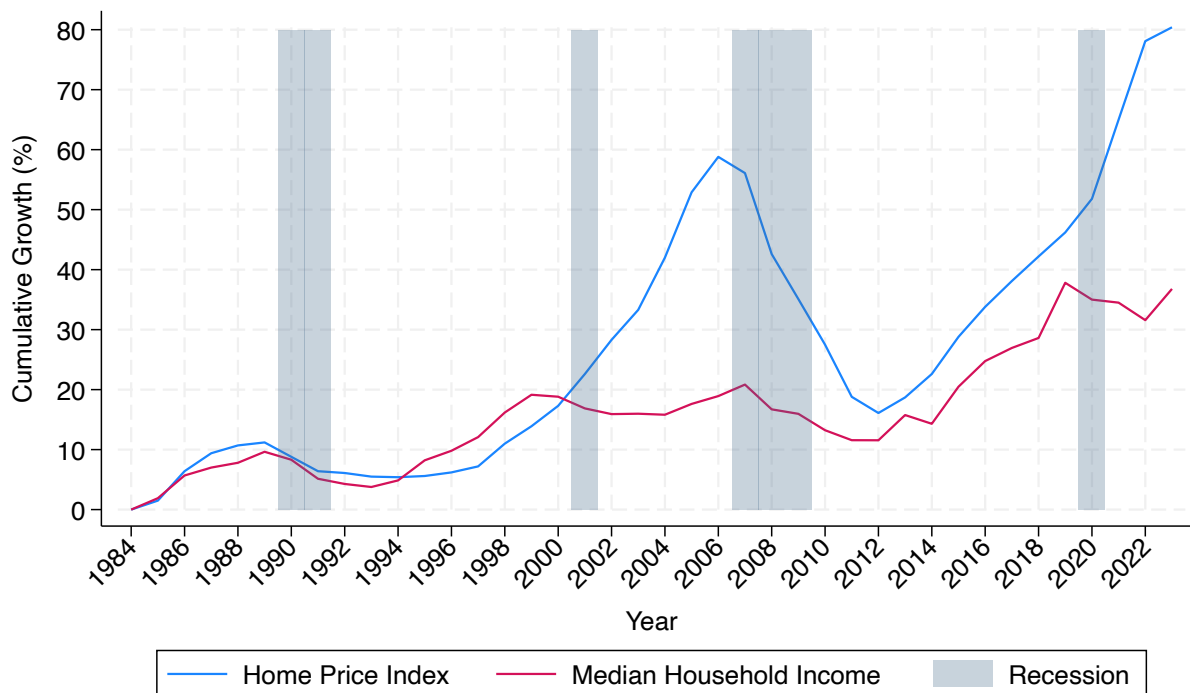
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1 Motivation

In the past 30 years, home prices have greatly out paced median household income (Figure 1.1), making homeownership and rents unaffordable for many young Americans (Quigley and Raphael (2004)). This has led many politicians to declare a “housing crisis.” However, there is significant heterogeneity in home prices across the country, with the Sun Belt and Midwest having lower prices, while the West Coast and Northeast have higher prices. If we compare this to a map of land use intensity (Figure 1.3), we see there is meaningful overlap with where housing prices are the greatest and where land use intensity is greatest, suggesting a relationship.

Figure 1.1: Real Changes in Housing Prices and Real Median Household Income Growth between 1984 to 2023.



Source: Federal Housing Finance Agency (HPI) and U.S. Census Bureau via FRED, Federal Reserve Bank of St. Louis.

The literature has found evidence for this relationship. The first national composite index of land use regulations was made by Gyourko, Saiz, and Summers (Gyourko et al. (2008))

with their Wharton Residential Land Use Regulatory Index (*WRLURI*). They sent out a survey with around 2,600 municipalities responding, capturing zoning restrictions, approval processes and delays, lot-size minimums, density constraints, open-space requirements, and local political influence in planning decisions into a composite index.

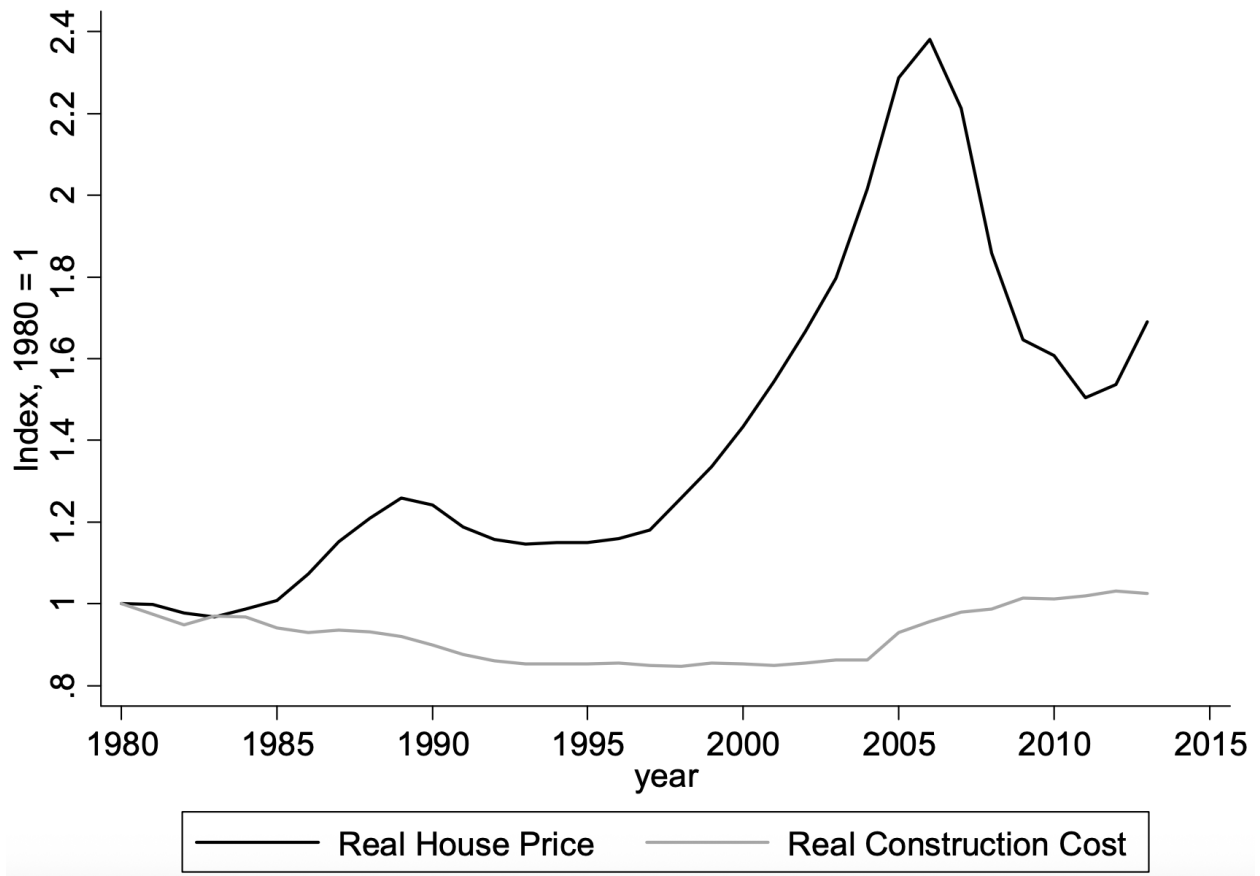
Saiz (2010) found that geographic constraints caused by steep slopes was positively correlated with the *WRLURI* suggesting that the scarcity of buildable land drives land management concerns. Another theory on the driver of land use regulations comes from Fischel (2001) with his Homevoter Hypothesis, which is the idea that homeowners use land use regulations as a tool to protect home values.

Gyourko and Molloy (2015) found that land use regulations increase the marginal cost of housing, which reduces construction and makes the housing market more inelastic. They found that home values have outpaced both construction costs and land values, and suggest that this difference in value is an implicit “regulatory tax” caused by land use regulations (Figure 1.2). They stressed that due to the endogenous nature of land use regulations (the high correlation between land use regulations and other attributes) and most papers being cross-sectional, most evidence in favor of land use regulations increasing home prices is associational. Reverse causality is a big concern since high prices may encourage homeowners to enact policies that protect home values.

With the updated *WRLURI* in 2018 by Gyourko, Hartley, and Krimmel (Gyourko et al. (2021)), changes in land use regulations over time could be measured after roughly a decade. They found that after the Great Recession, most municipalities had modest increases in land use intensity. To exploit this variation over time, in my undergraduate thesis¹, I constructed the $\Delta WRLURI$, a first difference measure of the index allowing for comparison of changes rather than levels. I was not the first to do this (Schoof (2021) and Wassmer and Williams (2021)), but I was the first to harmonize the two survey data sets and to examine the changes on the place (municipal) level instead of the metropolitan level, which averaged out significant

¹<https://economics.virginia.edu/sites/economics.as.virginia.edu/files/2025-05/Thoburn%2C%20Joseph.pdf>

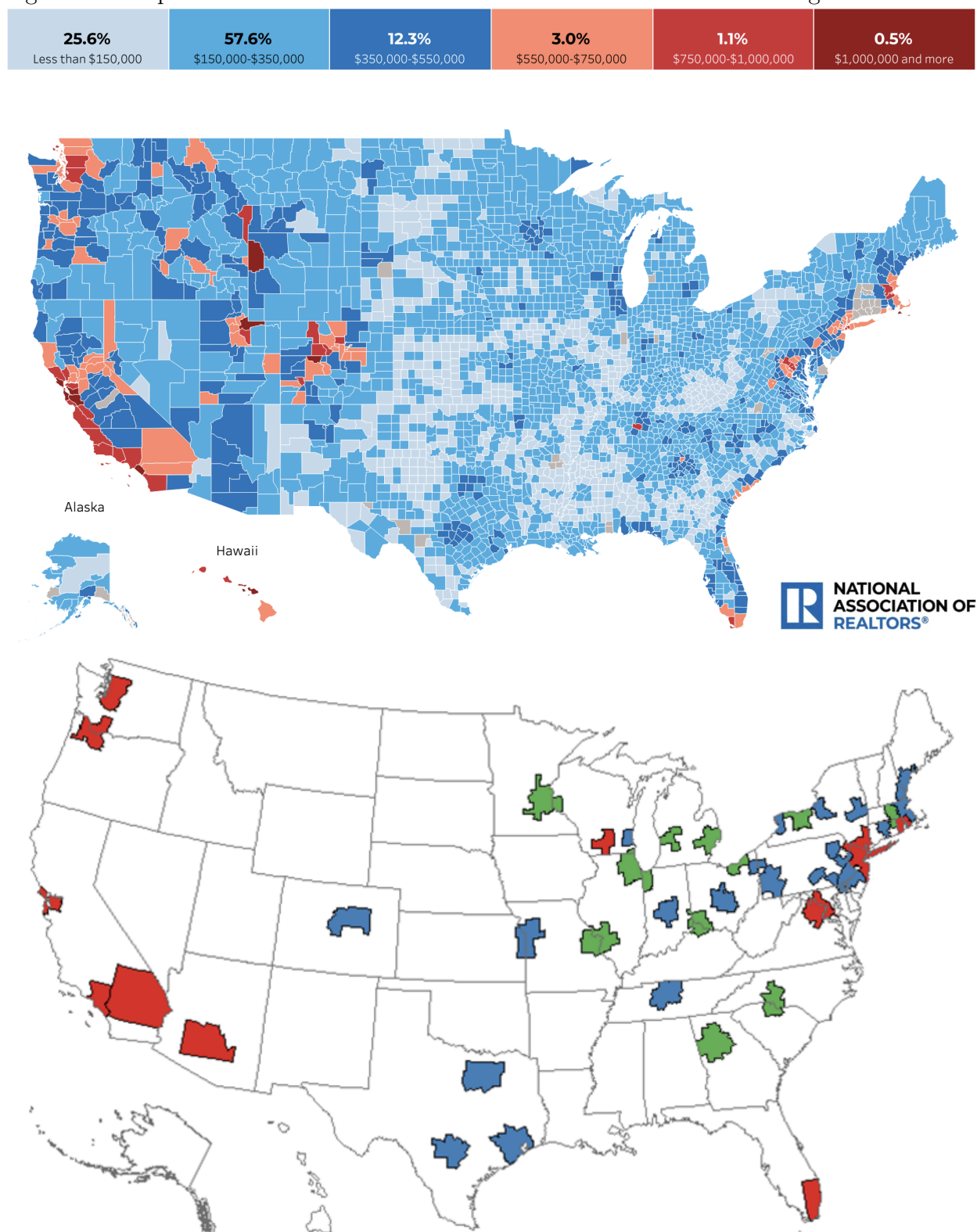
Figure 1.2: Real Construction Costs and House Prices Over Time.



Note: Construction costs are the cost of an Economy Quality home from R.S. Means Company deflated by the Consumer Price Index. House prices are the repeat-sales index published by CoreLogic deflated by the price index for personal consumption expenditures excluding housing services. The deflator was calculated by Gyourko & Molloy from data published by the Bureau of Economic Analysis.

local heterogeneity. Paradoxically, despite the vast literature connecting changes in land use regulations to changes on home prices, I found weak evidence in my models, which led me to return to the fundamentals.

Figure 1.3: Map of Median Home Prices in 2025 & Variation in Land Use Regulation in 2018.



Note: NAR Calculations of American Community Survey data; Federal Housing Finance Agency data; 2018 CBSA level Wharton Residential Land Use Regulatory Index; Red, top 11 values, Blue, middle 22 values, and Green, bottom 11 values.

2 The Theory

The kinked housing supply model comes from Glaeser and Gyourko (2005), but is standard in the literature. Housing is unlike other normal goods in that housing is a durable good, meaning that in the short run, housing construction takes time and depreciation is slow. This leads to a kinked supply curve, where supply is fixed (vertical) at H , the current housing stock, when P , the market price, is less than c , the minimum construction cost, and becomes upward sloping, when $P \geq c$. This is to say that developers will only build new houses if the market price is greater than the minimum cost of construction, allowing for profit. Thus $Q^s(P)$, the short run supply curve, is as follows:

$$Q_0^s(P) = \begin{cases} H, & \text{if } P < c, \\ H + s(P - c), & \text{if } P \geq c, \end{cases}$$

where $s > 0$ is the slope parameter.

Land use regulations affect supply through two channels. First, regulations on housing quality and impact fees (or proffers in Virginia), raise the marginal cost of construction, shifting the supply curve upward. Let τ represent this regulatory burden, making $c + \tau$ represent the new minimum construction cost. Second, land use regulations create physical constraints (e.g. density restrictions) and procedural constraints (e.g. approval delays). These regulations cause supply to become more inelastic, due to fewer housing substitutes (e.g. restrictions on manufactured homes) and through limiting developers' ability to respond to new demand, leading to a decrease in the slope parameter, that is, s becomes a function of γ , where γ is a measure of regulatory constraints on production (Gyourko and Molloy (2015)). Thus $Q^s(P)$ becomes:

$$Q_1^s(P) = \begin{cases} H, & \text{if } P < c + \tau, \\ H + s(\gamma)(P - c - \tau), & \text{if } P \geq c + \tau. \end{cases}$$

In high vacancy markets, where demand is weak relative to supply, $Q^d(P) = H$, the equilibrium lies on the vertical portion of supply. The impact of τ and γ are only felt when $P \geq c$, so the impact of land use regulations on the market price and quantity is not binding:

$$\frac{\partial P}{\partial \tau} = 0, \quad \frac{\partial Q}{\partial \tau} = 0, \quad \frac{\partial P}{\partial \gamma} = 0, \quad \frac{\partial Q}{\partial \gamma} = 0.$$

In low vacancy markets, where demand is strong relative to supply, $Q^d(P) > H$, the equilibrium lies above the kink and the impact of land use regulations on the market price and quantity is binding:

$$\frac{\partial P}{\partial \tau} > 0, \quad \frac{\partial Q}{\partial \tau} < 0, \quad \frac{\partial P}{\partial \gamma} > 0, \quad \frac{\partial Q}{\partial \gamma} < 0.$$

Figure 2.1 shows the two markets impacted by increases in land use regulations with different levels of market tightness.

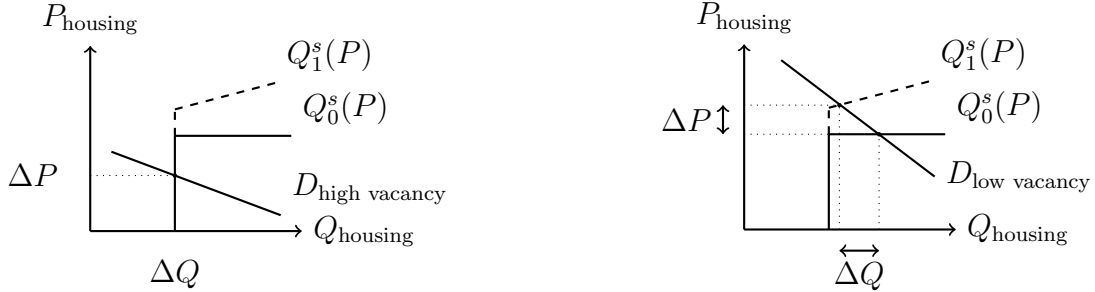
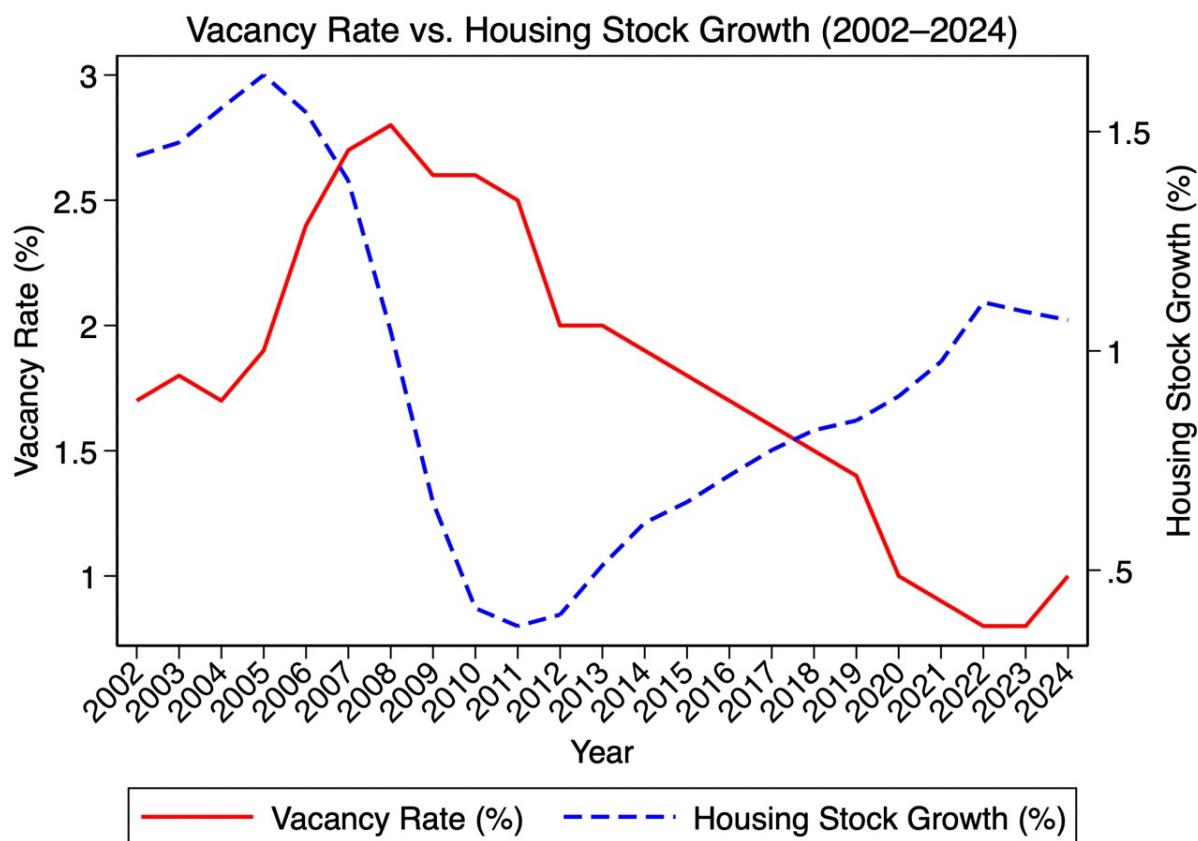


Figure 2.1: Two-panel diagram showing how a regulatory-induced shift from $Q_0^s(P)$ to a higher and more inelastic $Q_1^s(P)$ generates no change in prices or quantities in high-vacancy markets (left), but induces higher prices ($\Delta P > 0$) and lower quantities ($\Delta Q < 0$) in low-vacancy markets (right).

Why is vacancy used as a measure of market tightness? Strictly speaking, in the model described the vacancy rate would always be zero since the market would clear, but for this model to be useful we relax the assumptions of insignificant information and transaction costs, allowing for some friction in the market and market power for sellers. This allows for some flexibility in the share of unoccupied homes, which is what the vacancy rate measures.

If vacancy is high, then there is an excess of unoccupied homes, meaning sellers have less market power and have to lower prices, which decreases the incentive to build new homes. In low vacancy markets, unoccupied homes are scarce, meaning that sellers have the market power to raise prices, encouraging the construction of new homes. Thus the vacancy rate can inform us on the relative strength of supply to demand. Figure 2.2 shows the relationship between housing stock growth and the vacancy rate, and how new construction gets consumed, lowering the vacancy rate over time.

Figure 2.2: Vacancy Rate and Housing Stock Growth over time.



Source: U.S. Census Bureau and U.S. Department of Housing and Urban Development via FRED, Federal Reserve Bank of St. Louis.

3 The Model

3.1 Baseline Specification

To test the claim the market tightness should moderate the impact of land use regulations on housing prices, I first examine the base effect with the following OLS model:

$$\% \Delta HPI_i = \beta_0 + \beta_1 \Delta WRLURI_i + (\% \Delta X_i)' \beta_2 + Z_i^{(2006)'} \gamma + \alpha_i + \epsilon_i$$

Where $\% \Delta HPI_i$, is the real percentage change in home prices measured with the Federal Housing Finance Agency Home Price Index aggregated to the place (municipal) level from census tract data with the housing stock as a weight. This allows us to capture ΔP in our housing model. The Consumer Price Index for All Urban Consumers (CPI-U) was used to put the home price index in terms of 2006 dollars.

The Wharton Residential Land Use Regulatory Index ($WRLURI$) is a measure of municipal-level land use regulatory intensity standardized to have a mean of zero and a standard deviation of one. Surveys asking about zoning restrictions, approval processes and delays, lot-size minimums, density constraints, open-space requirements, and local political influence in planning decision were sent out in 2006 and 2018 with nearly 800 municipalities responding to both surveys, allowing us to take a first-difference as follows:

$$\Delta WRLURI_i = WRLURI_{i,2018} - WRLURI_{i,2006}$$

This allows us to measure changes in land use regulations over time, capturing the impact of the shift in our housing model from $Q_0^s(P)$ to $Q_1^s(P)$ with β_1 , which we expect to be positive but interpret with caution due to the endogenous nature of land use regulations. The survey was not one-to-one, so I created my own index ($WRLURI^{Adjusted}$) where I only kept questions that appeared in both surveys and used principal component analysis to get

the weights for the index. The main benefit of $\Delta WRLURI^{Adjusted}$ is that changes in the index can be interpreted as absolute changes whereas changes in the $\Delta WRLURI$ have to be interpreted as relative changes. See Appendix 5.2 for graphs.

$\% \Delta X_i$ is a vector of control variables that represent changes between 2006 and 2018, whereas $Z_i^{(2006)}$ is a vector of control variables that control for initial conditions in 2006 measured with the Census. These include the Area Ruggedness Scale, which captures geographic building limitations, and socioeconomic controls, which capture demand side factors (Table 5.1). α_i captures state and regional fixed effects, and I used clustered standard errors on the county level to help account for some spatial correlation.

Table 3.1 shows the baseline specifications. We see only marginal positive significance for β_1 for model (1), but overall the results suggests that $\Delta WRLURI$ is not a significant predictor of home price growth. However, this hides significant heterogeneity, which is getting averaged out.

Table 3.1: Effect of $\Delta WRLURI$ on $\% \Delta HPI$ at the Municipal Level

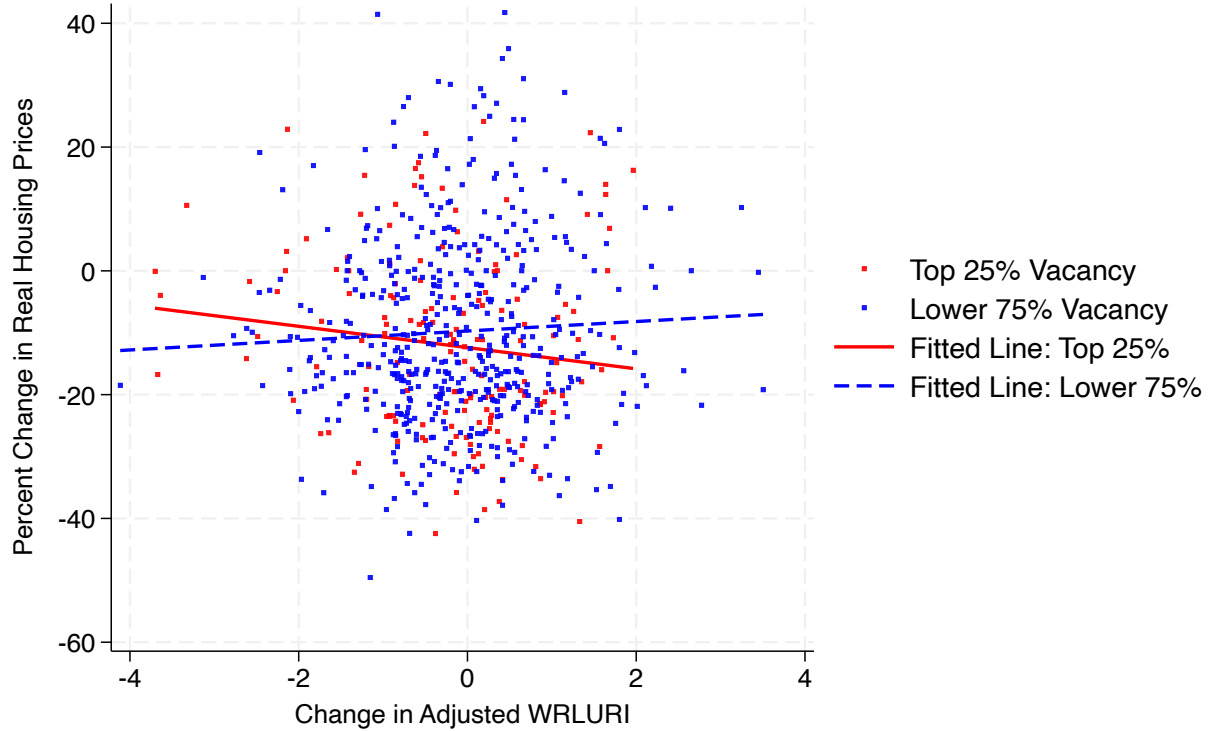
	(1)	(2)	(3)	(4)	(5)	(6)
	$\% \Delta HPI$	$\% \Delta HPI$	$\% \Delta HPI$	$\% \Delta HPI$	$\% \Delta HPI$	$\% \Delta HPI$
$\Delta WRLURI$ (original)	0.911*	0.077	0.161			
	(0.550)	(0.484)	(0.526)			
$\Delta WRLURI^{adjusted}$				0.240	-0.020	-0.202
				(0.502)	(0.385)	(0.455)
Socioeconomic Controls	X	X	X	X	X	X
Area Ruggedness Scale	X	X	X	X	X	X
State Fixed Effects		X			X	
Regional Fixed Effects			X			X
Observations	613	613	613	656	656	656
Adjusted R^2	0.217	0.605	0.371	0.204	0.600	0.374

Standard errors in parentheses. X indicates inclusion of the corresponding set of controls or fixed effects. Full regression results, including coefficients for all controls, are reported in Appendix Tables 5.2 and 5.3.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Looking at Figure 3.1, we see that the vacancy rate, as predicted by our housing model, moderates the impact. When municipalities are split into high vacancy (top quartile) and low vacancy (bottom quartiles) markets, the relationship between $\Delta WRLURI^{adjusted}$ and $\% \Delta HPI$ differ. In low vacancy markets, the slope is positive, consistent with the standard expectation that stricter regulations constrain supply and raise prices. In high vacancy markets, however, the slope is negative, suggesting that increases in regulatory stringency are associated with *declining* prices. This negative relationship is not causal, but likely reflects land use regulations being used to stabilize declining prices.

Figure 3.1: Real Change in Housing Prices and Change in Adjusted WRLURI.



Note: Real Change in Housing Prices and Change in Adjusted WRLURI. Higher-vacancy municipalities (top quartile) are shown in red, and lower-vacancy municipalities are shown in blue, with linear fitted lines for each group.

3.2 Modeling Heterogeneous Effects with the Vacancy Rate

We now add an interaction term between $\Delta WRLURI$ and $Vacancy$ to our baseline specification:

$$\begin{aligned} \% \Delta HPI_i = & \beta_0 + \beta_1 \Delta WRLURI_i + \beta_2 (\Delta WRLURI_i \times Vacancy_i) + \beta_3 Vacancy_i \\ & + (\% \Delta X_i)' \beta_4 + Z_i^{(2006)'} \gamma + \alpha_i + \epsilon_i. \end{aligned}$$

Such that

$$\frac{\partial (\% \Delta HPI_i)}{\partial (\Delta WRLURI_i)} = \beta_1 + \beta_2 \cdot Vacancy_i.$$

As before, we predict β_1 to be positive. We predict β_2 to be negative, since as the vacancy rate increases, supply becomes stronger relative to demand, thus moving demand toward the vertical portion of the supply curve where $\Delta WRLURI$ has no impact.

Table 3.2 shows the results, with β_1 consistently being positive and significant in the models and β_2 consistently being negative and significant in the models. Roughly speaking, a one standard-deviation increases in the intensity of land use regulations is associated with a 1% increase in real home prices with that impact decreasing by 0.1% for every 1% increase in the vacancy rate. Using model (5) at a vacancy rate of 9% (roughly the sample mean), the marginal association falls to

$$0.981 - (0.108 \times 9) = 0.009,$$

essentially zero. Above 9%, the total impact becomes negative, reflecting what we found in Figure 3.1. These results align with what our housing model predicts, but without an instrument variable to address the endogenous nature of land use regulations, the coefficient should not be interpreted as causal.

Table 3.2: Effect of $\Delta WRLURI \times Vacancy$ on $\% \Delta HPI$ at the Municipal Level

	(1)	(2)	(3)	(4)	(5)	(6)
	$\% \Delta HPI$	$\% \Delta HPI$	$\% \Delta HPI$	$\% \Delta HPI$	$\% \Delta HPI$	$\% \Delta HPI$
$\Delta WRLURI$ (original)	2.076*** (0.683)	1.074** (0.542)	1.194** (0.604)			
$\Delta WRLURI \times Vacancy$	-0.106*** (0.036)	-0.095*** (0.031)	-0.096** (0.041)			
$\Delta WRLURI^{adjusted}$				1.327* (0.697)	0.981* (0.592)	0.840 (0.634)
$\Delta WRLURI^{adjusted} \times Vacancy$				-0.115* (0.060)	-0.108** (0.051)	-0.110* (0.065)
Socioeconomic Controls	X	X	X	X	X	X
Area Ruggedness Scale	X	X	X	X	X	X
State Fixed Effects		X			X	
Regional Fixed Effects			X			X
Observations	613	613	613	656	656	656
Adjusted R^2	0.222	0.609	0.374	0.207	0.603	0.376

Standard errors in parentheses. X indicates inclusion of the corresponding set of controls or fixed effects. Full regression results, including coefficients for all controls, are reported in Appendix Tables 5.4 and 5.5

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4 Applications

This paper expands upon the literature by showing that vacancy rates moderate the impact of land use regulations on housing prices, reflecting the standard kinked supply housing model. This is important because it can provide insight into when it is appropriate for policy makers to use land use regulations. For example, where home prices are dropping, if land use regulations are used as a tool to stabilize prices, it won't be effective if vacancy rates are also high. Likewise, if housing affordability is a concern and vacancy rates are low, then increasing land use regulations will have a negative impact on affordability.

5 Appendix

5.1 Summary Statistics & Full Regression Outputs

Table 5.1: Summary Statistics

Variable	N	Mean	SD	Min	Max
Percent Change in Real HPI	659	-10.351	15.200	-49.549	41.737
Change in WRLURI (Old)	616	0.008	1.032	-3.992	3.097
Change in WRLURI (Adjusted)	659	-0.140	1.036	-4.115	3.503
Total Population (2006)	659	32,631	78,806	2,500	1,536,632
Percent Change in Population	659	8.196	14.926	-25.143	130.329
Median Household Income (2006)	659	54,401	25,309	16,025	234,846
Percent Change in Household Income	659	2.658	12.295	-33.479	71.828
Percent Urban (2006)	656	94.149	15.140	0	100
Percent College Educated (2006)	659	27.860	15.703	3.582	77.400
Percent Minority (2006)	659	17.783	15.574	0	86.260
Percent Unemployed (2006)	659	5.150	2.588	0	21.122
Percent in Poverty (2006)	659	12.200	8.153	1.058	44.947
Median Year Built (2006)	659	1971.621	14.031	1939	2003
Percent Vacant (2006)	659	9.354	6.930	0	71.087
Percent Homeowner (2006)	659	68.113	13.987	24.602	99.610
Percent Democrat (2004)	659	46.151	12.749	11.635	81.806

Table 5.2: Regulatory Changes and Housing Price Growth (Baseline WRLURI)

	(1)	(2)	(3)
	% ΔHPI	% ΔHPI	% ΔHPI
$\Delta WRLURI$ (original)	0.911*	0.077	0.161
	(0.550)	(0.484)	(0.526)
Total Population (2006)	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)
Percent Change in Population	0.222***	0.111***	0.188***
	(0.055)	(0.032)	(0.047)
Median Household Income (2006)	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)
Percent Change in Household Income	0.121**	0.098**	0.095*
	(0.054)	(0.042)	(0.051)
Percent Urban (2006)	0.045	0.022	0.091**
	(0.039)	(0.036)	(0.039)
Percent College Educated (2006)	0.259***	0.206***	0.230***
	(0.073)	(0.061)	(0.068)
Percent Minority (2006)	-0.020	-0.035	-0.114**
	(0.060)	(0.048)	(0.054)
Percent Unemployed (2006)	-0.989***	-0.010	-0.582**
	(0.276)	(0.217)	(0.269)
Percent in Poverty (2006)	0.272**	-0.293***	0.048
	(0.123)	(0.101)	(0.117)
Median Year Built (2006)	-0.142**	-0.128***	-0.130**
	(0.058)	(0.044)	(0.052)
Percent Vacant (2006)	-0.163	-0.097	-0.197*
	(0.108)	(0.088)	(0.114)
Percent Homeowner (2006)	0.038	-0.069	-0.015
	(0.089)	(0.067)	(0.080)
Percent Democrat (2004)	-0.338***	-0.046	-0.120
	(0.071)	(0.056)	(0.074)
Constant	275.779**	248.425***	250.140**
	(113.670)	(85.907)	(101.864)
ARS fixed effects	X	X	X
State fixed effects		X	
RSRegion fixed effects			X
Observations	613	613	613
Adj. R-squared	0.217	0.605	0.371

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5.3: Regulatory Changes and Housing Price Growth (Adjusted WRLURI)

	(4)	(5)	(6)
	% ΔHPI	% ΔHPI	% ΔHPI
$\Delta WRLURI^{Adjusted}$	0.240	-0.020	-0.202
	(0.502)	(0.385)	(0.455)
Total Population (2006)	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)
Percent Change in Population	0.205***	0.116***	0.180***
	(0.053)	(0.032)	(0.045)
Median Household Income (2006)	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)
Percent Change in Household Income	0.134**	0.103***	0.107**
	(0.054)	(0.038)	(0.048)
Percent Urban (2006)	0.048	0.008	0.078**
	(0.037)	(0.032)	(0.038)
Percent College Educated (2006)	0.193***	0.212***	0.178***
	(0.072)	(0.058)	(0.065)
Percent Minority (2006)	0.003	-0.009	-0.101**
	(0.060)	(0.047)	(0.051)
Percent Unemployed (2006)	-1.120***	-0.126	-0.696***
	(0.268)	(0.207)	(0.262)
Percent in Poverty (2006)	0.221*	-0.283***	0.010
	(0.123)	(0.101)	(0.111)
Median Year Built (2006)	-0.155***	-0.136***	-0.134***
	(0.054)	(0.043)	(0.050)
Percent Vacant (2006)	-0.166*	-0.112	-0.199*
	(0.100)	(0.085)	(0.110)
Percent Homeowner (2006)	0.020	-0.056	-0.039
	(0.081)	(0.062)	(0.071)
Percent Democrat (2004)	-0.339***	-0.044	-0.110
	(0.070)	(0.054)	(0.071)
Constant	303.679***	264.486***	260.874***
	(105.696)	(85.427)	(97.433)
ARS fixed effects	X	X	X
State fixed effects		X	
Regional fixed effects			X
Observations	656	656	656
Adj. R-squared	0.204	0.600	0.374

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5.4: Regulatory Changes, Vacancy, and Housing Price Growth (Baseline WRLURI)

	(1)	(2)	(3)
	$\% \Delta HPI$	$\% \Delta HPI$	$\% \Delta HPI$
$\Delta WRLURI$ (original)	2.076*** (0.683)	1.074** (0.542)	1.194** (0.604)
Percent Vacant (2006)	-0.222** (0.089)	-0.152** (0.074)	-0.256*** (0.098)
$\Delta WRLURI \times Vacancy$	-0.106*** (0.036)	-0.095*** (0.031)	-0.096** (0.041)
Total Population (2006)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Percent Change in Population	0.225*** (0.054)	0.114*** (0.032)	0.191*** (0.046)
Median Household Income (2006)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Percent Change in Household Income	0.114** (0.053)	0.093** (0.042)	0.091* (0.051)
Percent Urban (2006)	0.043 (0.039)	0.019 (0.036)	0.088** (0.039)
Percent College Educated (2006)	0.268*** (0.072)	0.211*** (0.059)	0.236*** (0.067)
Percent Minority (2006)	-0.022 (0.061)	-0.041 (0.048)	-0.117** (0.054)
Percent Unemployed (2006)	-0.956*** (0.279)	0.012 (0.217)	-0.550** (0.271)
Percent in Poverty (2006)	0.282** (0.124)	-0.281*** (0.100)	0.057 (0.118)
Median Year Built (2006)	-0.141** (0.058)	-0.133*** (0.043)	-0.133** (0.052)
Percent Homeowner (2006)	0.039 (0.090)	-0.071 (0.068)	-0.017 (0.081)
Percent Democrat (2004)	-0.341*** (0.071)	-0.048 (0.056)	-0.125* (0.074)
Constant	274.519** (113.103)	259.160*** (84.862)	255.659** (101.041)
ARS fixed effects	X	X	X
State fixed effects		X	
Regional fixed effects			X
Observations	613	613	613
Adj. R-squared	0.222	0.609	0.374

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5.5: Regulatory Changes, Vacancy, and Housing Price Growth (Adjusted WRLURI)

	(4)	(5)	(6)
	$\% \Delta HPI$	$\% \Delta HPI$	$\% \Delta HPI$
$\Delta WRLURI^{Adjusted}$	1.327*	0.981*	0.840
	(0.697)	(0.592)	(0.634)
$\Delta WRLURI^{Adjusted} \times Vacancy$	-0.115*	-0.108**	-0.110*
	(0.060)	(0.051)	(0.065)
Vacancy Rate (2006)	-0.212**	-0.157**	-0.246**
	(0.089)	(0.072)	(0.097)
Total Population (2006)	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)
Percent Change in Population	0.206***	0.118***	0.181***
	(0.053)	(0.031)	(0.045)
Median Household Income (2006)	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)
Percent Change in Household Income	0.132**	0.102***	0.107**
	(0.053)	(0.038)	(0.048)
Percent Urban (2006)	0.050	0.008	0.078**
	(0.037)	(0.032)	(0.039)
Percent College Educated (2006)	0.196***	0.214***	0.180***
	(0.071)	(0.057)	(0.064)
Percent Minority (2006)	-0.001	-0.017	-0.103**
	(0.060)	(0.048)	(0.051)
Percent Unemployed (2006)	-1.085***	-0.096	-0.661**
	(0.269)	(0.208)	(0.262)
Percent in Poverty (2006)	0.213*	-0.290***	-0.002
	(0.123)	(0.100)	(0.111)
Median Year Built (2006)	-0.153***	-0.137***	-0.134***
	(0.054)	(0.043)	(0.049)
Percent Homeowner (2006)	0.013	-0.064	-0.049
	(0.081)	(0.062)	(0.072)
Percent Democrat (2004)	-0.341***	-0.043	-0.113
	(0.070)	(0.054)	(0.071)
Constant	301.212***	269.362***	262.874***
	(105.384)	(84.882)	(97.000)
ARS fixed effects	X	X	X
State fixed effects		X	
Regional fixed effects			X
Observations	656	656	656
Adj. R-squared	0.207	0.603	0.376

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.2 $WRLURI^{Adjusted}$

Table 5.6: Comparison of Factor Loadings Across Years

Subindex	2006	2018	Pooled
LPPI	0.3505	0.4404	0.4212
SPII	0.2262	0.3391	0.2761
CII	0.3430	0.3378	0.3523
LPAI	0.4269	0.2761	0.3347
LZAI	0.2373	0.1768	0.1903
SRI	0.1548	0.1012	0.1289
DRI	0.3067	0.1038	0.1668
OSI	0.3683	0.3731	0.3847
EI	0.2094	0.4366	0.3636
AHI	0.3560	0.3233	0.3549
ADI	0.2133	0.1189	0.1535

I used principal component analysis (PCA) on a pooled sample from 2006 and 2018, so I could make the interpretation straightforward. For example, if I had used PCA on both years separately and calculated $\Delta WRLURI$, a large change would not necessarily mean regulation or deregulation. For example, a municipality could have added new regulations but if the rest of the country had added more regulations on average, then their $\Delta WRLURI$ would have been negative, which is not economically useful.

The weights from the PCA are all positive, suggesting that the subindices all contribute in the same direction as expected (regulation increases total regulatory intensity) and the weights are relatively stable regardless of the sample. Looking at the factor weights from the original papers, LPPI, SPII, and CII are also near the top. Full detail, though not as clear as it should be, on how I summarized the survey responses can be found in the appendix of my original thesis². The graphs below are new and did not make it into the original paper.

²<https://economics.virginia.edu/sites/economics.as.virginia.edu/files/2025-05/Thoburn%20C%20Joseph.pdf>

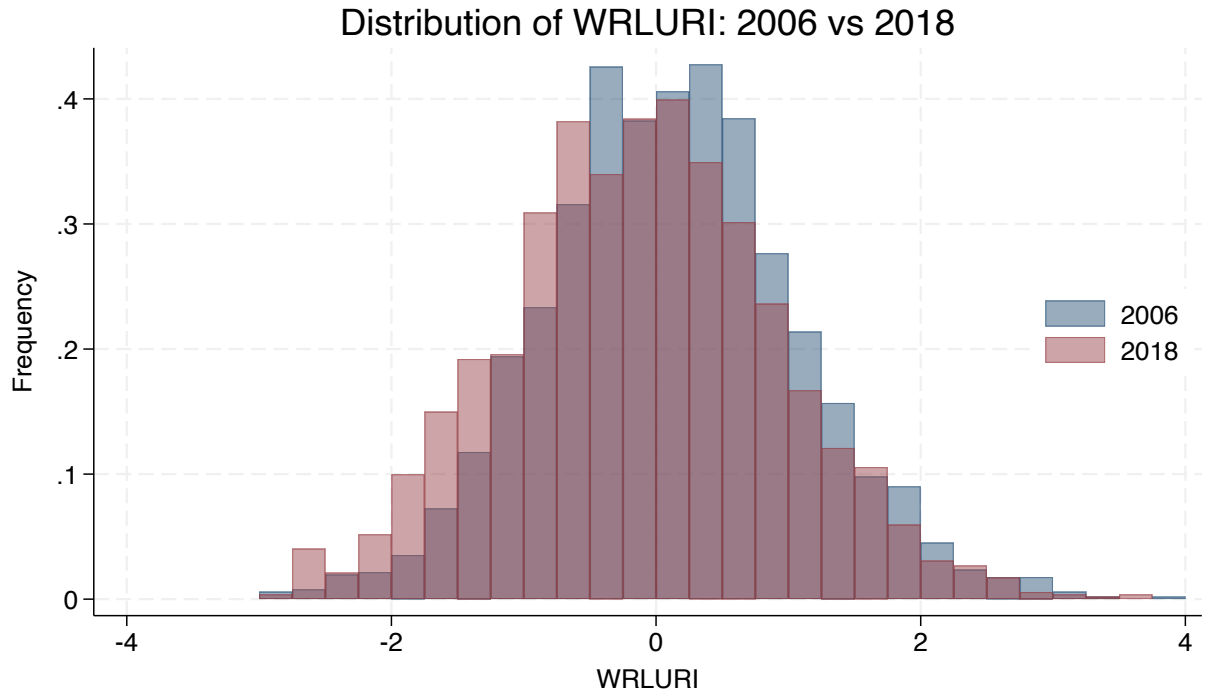


Figure 5.1: Distribution of $WRLURI^{Adjusted}$ by Year (4,122 obs); full sample.

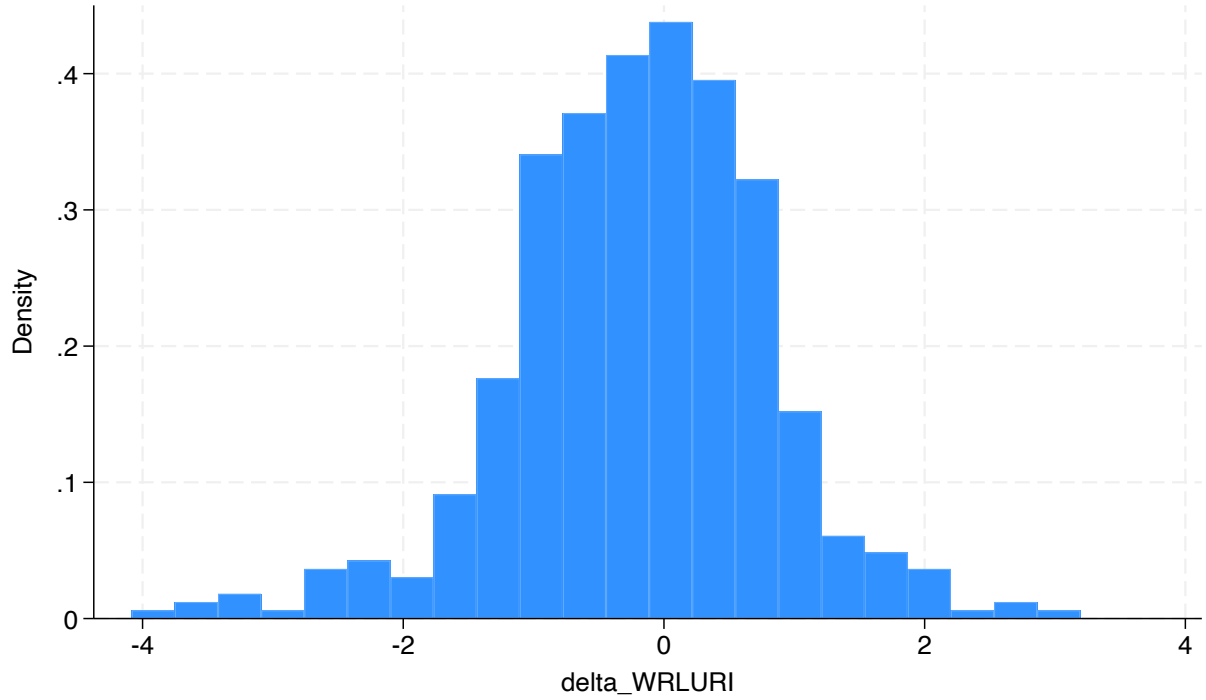
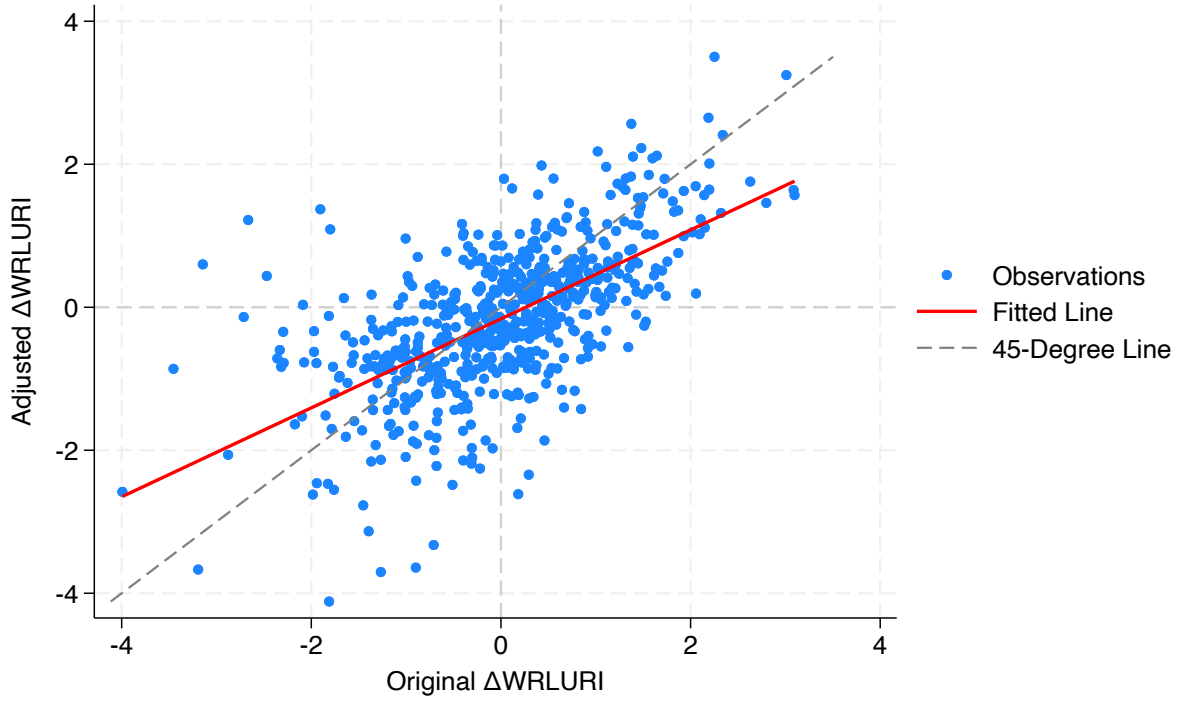


Figure 5.2: Distribution of $\Delta WRLURI^{Adjusted}$ (497 obs); this sample is smaller since the larger sample of 659 includes linear predictions for at most one subindex component.

Figure 5.3: Relationship Between Original and Adjusted $\Delta WRLURI$.



Note: The $\Delta WRLURI^{Adjusted}$ is recalculated using only the questions consistently measured in both survey years (2006 and 2018). The fitted regression line (solid red) illustrates the relationship between the original and adjusted indices, while the dashed 45-degree line shows points of exact equivalence.

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